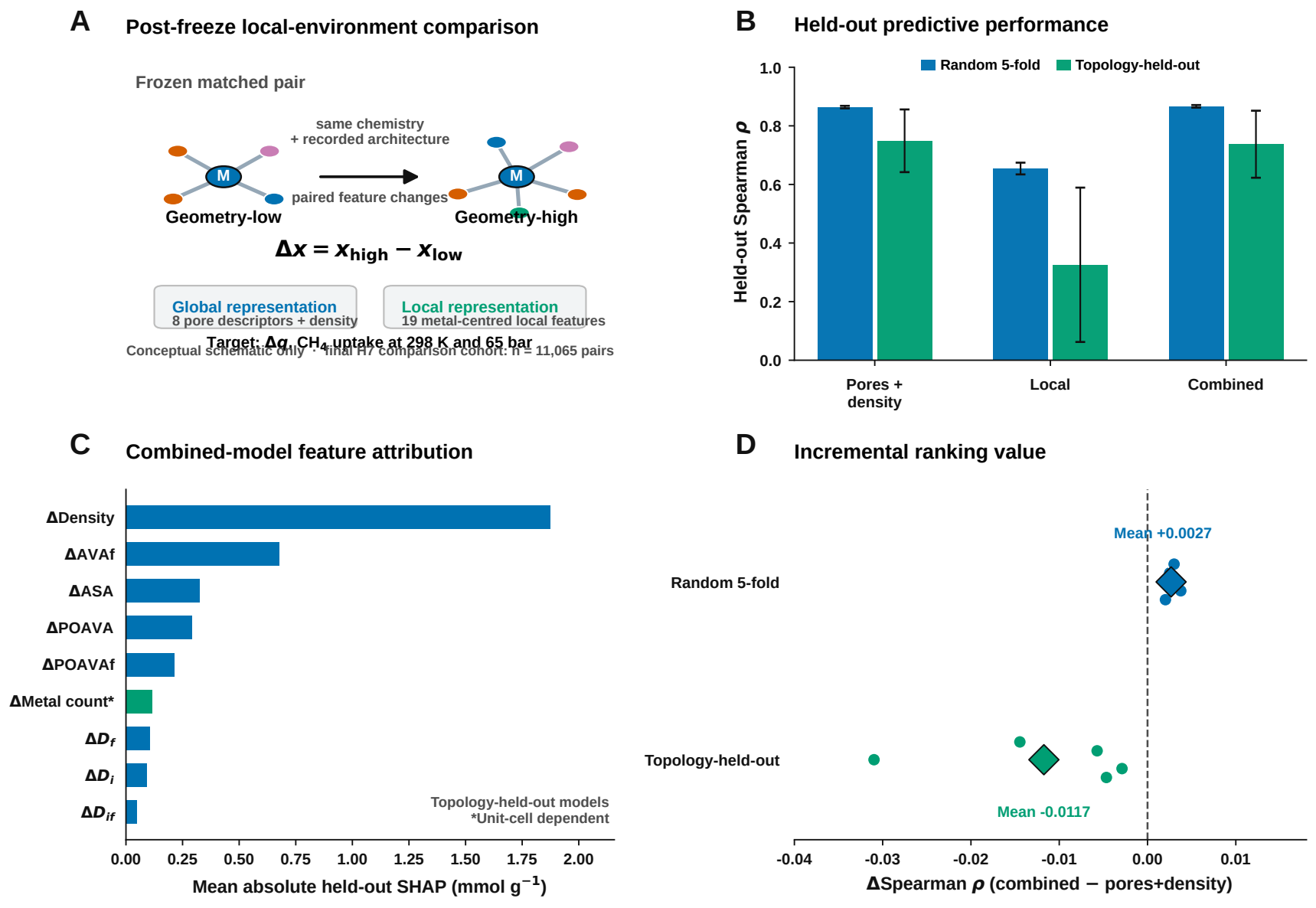
A
Post-freeze local-environment comparison
Frozen matched pair
same chemistry
+ recorded architecture
paired feature changes
M
M
Geometry-low
Geometry-high
$\Delta x = x_{high} - x_{low}$
Global representation
8 pore descriptors + density
Local representation
19 metal-centred local features
Target: Δq, CH_4 uptake at 298 K and 65 bar
Conceptual schematic only · final H7 comparison cohort: n = 11,065 pairs
B
Held-out predictive performance
Random 5-fold
Topology-held-out
Held-out Spearman ρ
1.0
0.8
0.6
0.4
0.2
0.0
Pores + density
Local
Combined
C
Combined-model feature attribution
ΔDensity
ΔAVAf
ΔASA
ΔPOAVA
ΔPOAVAf
ΔMetal count*
ΔD_f
ΔD_i
ΔD_{if}
Topology-held-out models
*Unit-cell dependent
0.00
0.25
0.50
0.75
1.00
1.25
1.50
1.75
2.00
Mean absolute held-out SHAP (mmol g^{-1})
D
Incremental ranking value
Mean +0.0027
Random 5-fold
Topology-held-out
Mean -0.0117
-0.04
-0.03
-0.02
-0.01
0.00
0.01
ΔSpearman ρ (combined − pores+density)